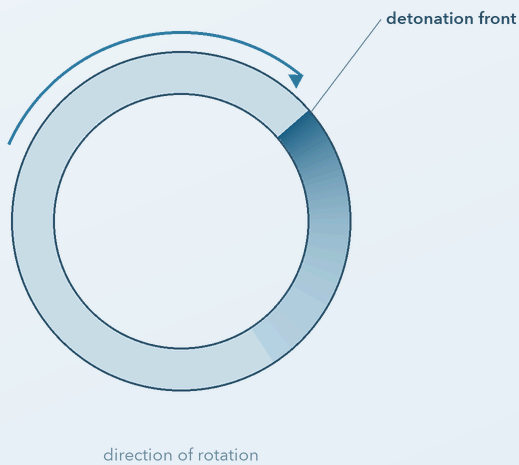


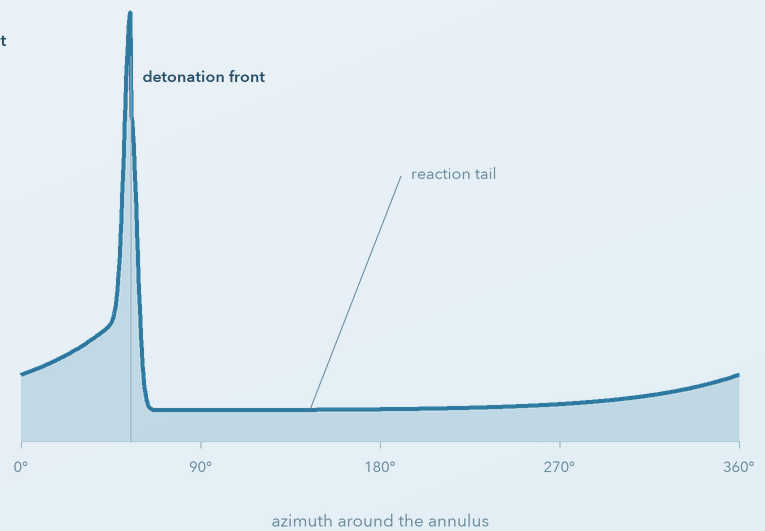
Resonance

A student-led rotating detonation rocket engine program at UIUC.

Annular pressure field



Pressure around one revolution



Resonance Mk1 is a 2 kN, water-cooled LOX-methane engine designed for a five-to-ten-second first campaign at UIUC's SPIRIT facility.

A path to practical propulsion

The first campaign focuses on sustained detonation, measured thrust, and reusable hardware. The longer path leads toward upper-stage and in-space propulsion.

With Resonance, we aim to turn rotating-detonation physics into durable engine hardware by progressing from sustained detonation and calibrated thrust to cooling, repeatability, and later restart.

Upper stages

Compact hardware can reduce packaging and dry-mass burdens.

Orbital transfer

Efficient cryogenic propulsion can support movement between orbits.

Landers

Short, reusable engines can open new vehicle layouts.

Deeper-space missions

Long missions reward durability, thermal control, and restart.

The first milestone

Building on public work from NASA and Purdue: a clear detonation signature, calibrated thrust, and at least five seconds of stable operation.

Project lead



Syed Aabis Akhtar

Aerospace Engineering, UIUC
Engine Fluids, Liquid Rocketry at Illinois

Aabis has built products and technical systems since high school, then moved into propulsion at UIUC. At Liquid Rocketry at Illinois, he has built feed-system hardware from P&IDs and supported pressure checks, cold flows, ignition testing, and Overture Mk1 hotfire operations.

Built early

Co-founded StellarSOL in tenth grade and raised \$50,000 in equity-free grants and prizes.

Learned fast

Learned Swift and built AsymmED in four days without owning a Mac; won Apple's 2023 Swift Student Challenge.

Shipped systems

Built Solathon to 150,000+ downloads, then moved into feed-system hardware and hotfire operations.



Overture Mk1 hotfire, May 2026. Aabis supported fluids and test-stand operations.

Resonance Mk1 architecture

Resonance Mk1 is a pressure-fed, water-cooled, predetonated liquid-oxygen and liquid-methane RDRE. One replaceable injector head serves three cooled barrels so combustor length remains the experiment.

2.0 kN

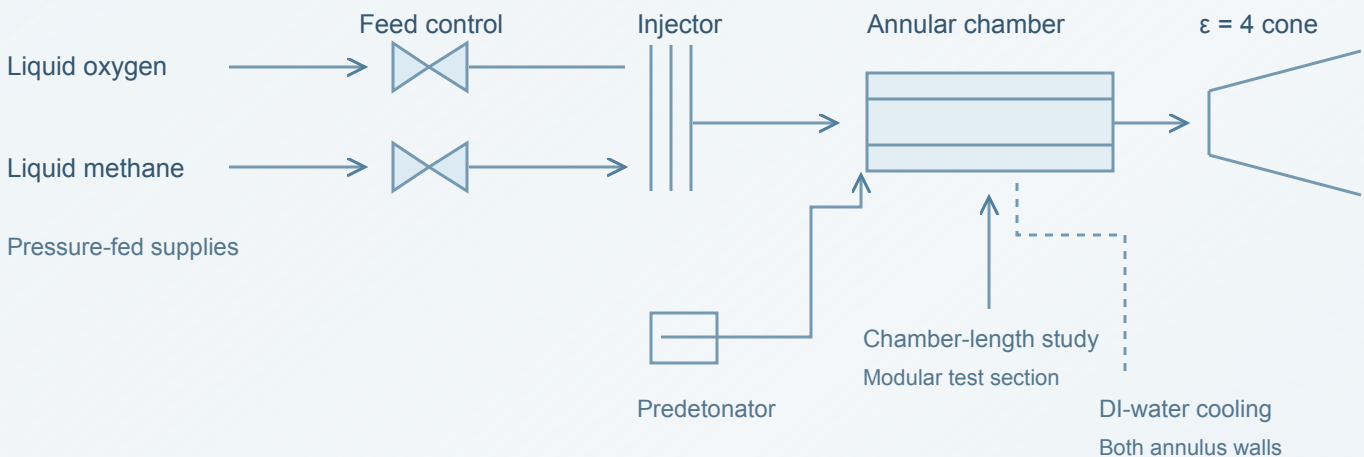
Nominal; 1.5-2.5 kN operating envelope

350 psia

Centre chamber pressure; 250 psia commissioning

85 × 5 mm

Centre annular gas path



The experiment

Three cooled barrels provide L' values of 1.5, 2.0, and 2.5 in. The first article is 2.0 in; separate barrels avoid spacer joints in the hot gas path.

Build approach

LPBF GRCo-42 with computed-tomography inspection is preferred if a qualified supplier is available. A CNC OFHC calorimeter is the backup route.

Resonance Mk1 at a glance

A compact, pressure-fed engine sized for a meaningful first campaign at SPIRIT.

2.0 kN

Nominal thrust

350 psia

Chamber pressure

LOX / CH₄

Propellants

5–10 s

First campaign

85 × 5 mm

Annular gas path

0.878 kg/s

Nominal total flow

Combustor

Three cooled barrels at 1.5, 2.0, and 2.5 inches. The 2.0-inch barrel fires first.

Cooling

DI water on both annulus walls, sized around a roughly 47 gpm center estimate.

Ignition

An independently qualified predetonator initiates the Mk1 engine.

Nozzle

A 29.1 mm throat and area-ratio-four conical nozzle form the current baseline.

First success

A rotating-detonation signature, calibrated thrust, and at least five seconds of thermal survival.

Why the centre geometry is 85 × 5 mm

The selected annulus keeps contraction inside the 1.6–2.0 design range while leaving room for cooling jackets and instrumentation.

1.89

Contraction ratio at the 85 × 5 mm centre

699

kg/m²s annulus mass flux at 0.878 kg/s

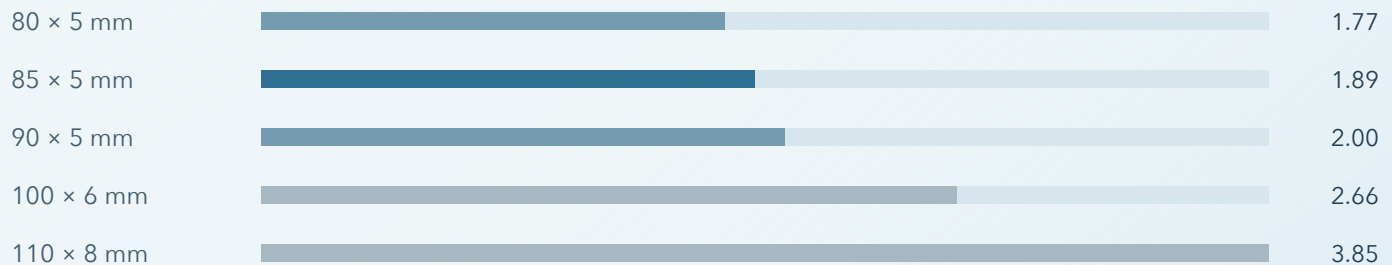
29.1 mm

Thermodynamic throat diameter

5.0–5.5 in

Working hardware outer diameter

Contraction ratio against the 29.1 mm throat



The earlier 100–110 mm study box sits well above the selected contraction range.

**Performance
estimate**

**350 psia · O/F
3.25**

3396 K

Chamber
temperature

1831 m/s

Characteristic
velocity

273 s

True one-
atmosphere Isp

0.878

kg/s

0.672 LOX + 0.207
methane kg/s

The first research question

How does combustor length change detonation stability, thrust, and thermal loading?

1.5 in	2.0 in	2.5 in
Shortest cooled barrel	First article	Longest cooled barrel
Hold approximately constant	Injector head, 85 × 5 mm annulus, throat family, propellants, operating point, and instrumentation.	
Compare between configurations	Wave stability, thrust, chamber pressure, mass flow, and thermal response.	

Evidence needed from each hotfire

Thrust	LOX mass flow	Methane mass flow
Chamber pressure	Thermal response	Dynamic pressure
Wave frequency	Wave count and direction	Mode switching

Testing at UIUC

SPIRIT

The new testing facility at the University of Illinois Urbana-Champaign.

Resonance is sized to fit comfortably inside the facility envelope.

10 kN

Maximum thrust

The 2 kN engine target is one-fifth of the stated limit.



500 psi

Maximum chamber pressure

The 350 psia center remains below the stated chamber limit.

12 in

Maximum diameter

A 5.0–5.5 inch engine leaves room for cooling, plumbing, and instrumentation.



An existing university test facility keeps the first campaign focused on engine hardware, instrumentation, and iteration.

Working plan to first hotfire

The first approved hotfire is targeted for Spring 2027.

**Sep-Oct
2026**

Requirements and team

Define requirements, recruit the core build team, and complete the campaign safety plan.

**Nov-Dec
2026**

Preliminary design

Freeze the operating point, injector concept, cooling approach, and manufacturing plan.

Jan-Feb 2027

Detailed design and procurement

Release drawings, select vendors, and order long-lead hardware.

**Mar-Apr
2027**

Build and integration

Complete inspection, pressure checks, instrumentation, water flow, and cold flow.

Spring 2027

First hotfire window

Begin the approved campaign and work toward at least five seconds of sustained operation.

Resonance Mk1

Stable detonation, chamber-length study, and measurable thrust.

Resonance Mk2

Cooling, duration, repeated hotfires, and durability.

Resonance Mk3

Compact ignition, controlled shutdown, restart, and repeatability.

What the next year unlocks

The first year turns the present architecture into inspected hardware, an instrumented hotfire campaign, and evidence for the next engine.

Build Mk1

Manufacture and inspect the injector, hot section, nozzle, and three cooled barrels.

Commission the system

Qualify the predetonator, integrate feed and cooling hardware, and commission at 250 psia.

Run the experiment

Fire three chamber lengths and compare detonation stability, thrust, and thermal response.

Make the work legible

Share approved test media, campaign briefings, and a concise final technical report.

Ways to help

Cash, manufacturing, inspection, sensors, valves, materials, and engineering time can each map to a defined part of the build or campaign.

Fund the first campaign

We are raising cash and in-kind support for one year of hardware, instrumentation, integration, and test work required to reach the first sustained campaign at SPIRIT.

\$75,000

Target for Mk1
and its first campaign

The campaign includes the hot section, feed and cooling hardware, a predetonator, instrumentation, three cooled barrels, and spares.

Planned use of funds

GRCop-42 or OFHC hot section, injector head, finish machining, CT or inspection, and three barrels	\$15k-\$30k
Feed hardware, valves, controls, and calibrated flow metering	\$10k-\$25k
Four high-frequency pressure stations, thrust, thermal instrumentation, and DAQ	\$10k-\$25k
DI-water cooling loop, predetonator qualification, facility integration, and campaign contingency	\$15k-\$20k

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